

## Albumin/Creatinine Ratio, Random Urine

### Test Details

#### METHODOLOGY

**Albumin:** immunoturbidimetric; **Creatinine:** kinetic

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#### RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

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### Test Includes

Albumin, random urine; creatinine; albumin:creatinine ratio

### Use

Measurement of albumin levels in urine below the detection level of urine dipsticks. This test is useful in the management of patients with relatively early diabetes mellitus to assist in avoiding or delaying the onset of diabetic renal disease.

## Limitations

Because of the inherent day-to-day variability of albumin excretion into the urine, two of three albumin levels measured within a three-month to six-month period should be abnormal before considering a patient to have crossed a diagnostic threshold.<sup>1</sup> Physical exercise during the previous 24 hours can cause a transient elevation in albumin. Other variables, including infection, fever, congestive heart failure, marked hyperglycemia, and marked hypertension, can result in increased albumin levels.

## Custom Additional Information

Albumin accounts for approximately 50% of the protein in plasma.<sup>2</sup> The kidney works to prevent the loss of albumin into the urine through active resorption, but a small amount of albumin can be measured in urine of individuals with normal renal function.

The prognostic value of consistently elevated albumin levels is particularly well established in diabetic patients.<sup>1</sup> Renal disease is a common microvascular complication of diabetes. Without specific interventions, 80% of type I diabetics with repeatedly elevated albumin levels will go on to end-stage renal disease. Twenty percent to 40% of type II diabetics with sustained albuminuria will progress to overt nephropathy.

The American Diabetes Association (ADA) recommends that routine urinalysis be performed annually on adults with diabetes.<sup>1</sup> If the urinalysis is negative for protein, albumin measurement is recommended. The ADA also recommends annual screening of children beginning at puberty or after five years of disease duration. The reference intervals stated above reflect the diagnostic criteria prescribed by the ADA.<sup>1</sup>

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# Specimen Requirements

## Specimen

Urine (random)

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## Volume

10 mL

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## Minimum Volume

2 mL (**Note:** This volume does **not** allow for repeat testing.)

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## Container

Plastic urine container

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## Collection Instructions

Collect random urine without preservatives. pH must be 4 to 8.

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## Stability Requirements

| Temperature        | Period    |
|--------------------|-----------|
| Room temperature   | 7 days    |
| Refrigerated       | 14 days   |
| Frozen             | 14 days   |
| Freeze/thaw cycles | Stable x3 |

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## Reference Range

- Normal: 0–29 mg/g creatinine

- Moderately increased: 30–300 mg/g creatinine
  - Severely increased: >300 mg/g creatinine
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## Storage Instructions

Room temperature

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## Causes for Rejection

Bloody specimen

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## References

Metcalf P, Baker J, Scott A, Wild C, Scragg R, Dryson E. Albuminuria in people at least 40 years old: Effect of obesity, hypertension, and hyperlipidemia. *Clin Chem*. 1992 Sep; 38(9):1802-1808. PubMed 1526018 (<http://www.ncbi.nlm.nih.gov/pubmed/1526018>)

## Footnotes

1. American Diabetes Association. Position Statement: Diabetic Nephropathy. *Diabetes Care*. 1997; 20(50):S24-S27.
2. Sacks DB. Carbohydrates. In: Burtis CA, Ashwood ER, eds. *Tietz Textbook of Clinical Chemistry*. 3rd ed. Philadelphia, Pa: WB Saunders Co; 1999: 750-808.